Day 037 | Unit 03 | Teacher plan

Integers and Rational Numbers

Interpret Order in Context

Standards: 6.NS.C.7a, 6.NS.C.7b, 6.NS.C.5

Learning target: I can interpret Order in Context, and explain and check my reasoning.

enVision Topic 2: Integers and Rational Numbers | Spectrum companion Chapter 4

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. A number farther right is greater. Among negative numbers, the value closer to zero is greater. Use a common denominator or familiar benchmarks to compare negative fractions. Choose a reference point for zero. Opposite directions or values use opposite signs: above sea level is positive and below sea level is negative. Zero means at the chosen reference, not always nothing.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 22, 24

Day 037 | Unit 03 | Lesson

Integers and Rational Numbers

Interpret Order in Context

Standards: 6.NS.C.7a, 6.NS.C.7b, 6.NS.C.5

Learning target: I can interpret Order in Context, and explain and check my reasoning.

enVision Topic 2: Integers and Rational Numbers | Spectrum companion Chapter 4

01 / Learn

A number farther right is greater. Among negative numbers, the value closer to zero is greater. Use a common denominator or familiar benchmarks to compare negative fractions. Choose a reference point for zero. Opposite directions or values use opposite signs: above sea level is positive and below sea level is negative. Zero means at the chosen reference, not always nothing.

Worked example 1

Compare -7 ___ -6 using <, >, or =. Sketch a number line to justify.

1. A number farther right is greater.

2. Among negative numbers, the value closer to zero is greater.

3. -7 < -6

4. greater numbers lie farther right.

5. State the result in the form requested: <.

Answer: <

Worked example 2

A sensor is 5 meters below a reference point. Select all correct interpretations.
A. Its signed position is +5 m.
B. A reading of 0 m means at the reference point.
C. The opposite position is 5 m below the reference.
D. Its signed position is -5 m.

1. Choose a reference point for zero.

2. Below the chosen zero is negative, so the position is -5.

3. Zero identifies the reference

4. the opposite is 5 m above it.

Answer: B, D - A reading of 0 m means at the reference point.; Its signed position is -5 m.

Day 037 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Compare -5 ___ -2 using <, >, or =. Sketch a number line to justify.

02 / Constructed response

A marker is 12 m below sea level. Write its signed elevation and explain what elevation 0 means.

03 / Constructed response

Compare -3 ___ -4 using <, >, or =. Sketch a number line to justify.

04 / Constructed response

A marker is 32 m below sea level. Write its signed elevation and explain what elevation 0 means.

05 / Constructed response

Compare -4 ___ -1 using <, >, or =. Sketch a number line to justify.

06 / Constructed response

A marker is 8 m below sea level. Write its signed elevation and explain what elevation 0 means.

07 / Constructed response

Temperatures are -1/2°C and -1/5°C. Write an inequality comparing them and explain which is warmer.

08 / Multiple select

A sensor is 3 meters below a reference point. Select all correct interpretations.
A. Its signed position is -3 m.
B. A reading of 0 m means at the reference point.
C. The opposite position is 3 m below the reference.
D. Its signed position is +3 m.

09 / Multiple select

Temperatures are -6 °C and -4 °C. Select all true comparisons.
A. -6 is warmer than -4.
B. -6 < -4
C. -4 lies to the right of -6 on a number line.
D. -6 > 0

10 / Constructed response

A marker is 42 m below sea level. Write its signed elevation and explain what elevation 0 means. Write the elevation of the point equally far on the opposite side of sea level.

Day 037 | Unit 03 | Answer key

Integers and Rational Numbers

Interpret Order in Context

Standards: 6.NS.C.7a, 6.NS.C.7b, 6.NS.C.5

Learning target: I can interpret Order in Context, and explain and check my reasoning.

1. <

-5 < -2; greater numbers lie farther right.

6.NS.C.7a, 6.NS.C.7b | Constructed response

2. -12 m; 0 means at sea level

Sea level is the reference. Below is negative and above is positive.

6.NS.C.5 | Constructed response

3. >

-3 > -4; greater numbers lie farther right.

6.NS.C.7a, 6.NS.C.7b | Constructed response

4. -32 m; 0 means at sea level

Sea level is the reference. Below is negative and above is positive.

6.NS.C.5 | Constructed response

5. <

-4 < -1; greater numbers lie farther right.

6.NS.C.7a, 6.NS.C.7b | Constructed response

6. -8 m; 0 means at sea level

Sea level is the reference. Below is negative and above is positive.

6.NS.C.5 | Constructed response

7. -1/2 < -1/5; -1/5°C is warmer

The greater value is farther right on a number line, even when both are negative.

6.NS.C.7a, 6.NS.C.7b | Constructed response

8. A, B - Its signed position is -3 m.; A reading of 0 m means at the reference point.

Below the chosen zero is negative, so the position is -3. Zero identifies the reference; the opposite is 3 m above it.

6.NS.C.5 | Multiple select

9. B, C - -6 < -4; -4 lies to the right of -6 on a number line.

Among negatives, the number closer to zero is greater. Thus -4 is warmer and is farther right.

6.NS.C.7a, 6.NS.C.7b | Multiple select

10. -42 m; 0 means at sea level; opposite +42 m

Sea level is the reference. Below is negative and above is positive.

6.NS.C.5 | Constructed response